

PARKER HUNT

Chestnut Hill, MA | huntpb@bc.edu | parker-hunt.github.io
Department of Physics & Schiller Institute for Integrated Science and Society
Boston College

EDUCATION

Boston College | Chestnut Hill, MA

Ph.D. Candidate in Physics (Expected Graduation: Summer 2027)

- Research Specialization: Atmospheric Dynamics & Idealized Climate Modeling
- Advisor: Dr. Yi Ming

The Ohio State University | Columbus, OH

B.S. in Physics with Minor in Mathematics (May 2020)

- Honors: *Magna Cum Laude*, with Research Distinction in Physics
- GPA: 3.85 / 4.00 (*Physics Major GPA: 3.90 / 4.00*)

PUBLICATIONS & MANUSCRIPTS IN PREPARATION

1. **Hunt, P.**, Lin, P., Kim, S., MacDonald, C. G., & Ming, Y. (2026). Time-mean tropical circulation in an idealized moist GCM: Dependence on Zonally Asymmetric Forcings. (*Submitted to Journal of Climate*).
2. **Hunt, P.**, Lin, P., Kim, S., MacDonald, C. G., & Ming, Y. (2026). Time-mean tropical circulation in an idealized moist GCM: Response to CO₂ forcings. (*In preparation*).
3. Kim, S., **Hunt, P.**, & Ming, Y. (2026). Zonal Structure of Warming-Induced Tropical Circulation Changes. (*Submitted to Geophysical Research Letters*).
4. Landry, K., Zhou, Y., Ruan, X., Chen, J.-J., Zhai, R., Kong, T., Lueders-Dumont, J., Dong, S., Osorio-Rodriguez, D., Pavia, F. J., Subhas, A. V., Adkins, J. F., Berelson, W. M., Peng, X. N., **Hunt, P.**, Tian, H., Ming, Y., & Wang, X. T. (2026). Thickening of oxygen-deficient waters in the tropical Pacific. (*In preparation*).
5. **Hunt, P.**, Lin, P., & Ming, Y. (2026). Time-mean Tropical Circulation in an Idealized Moist GCM: Divergent Circulation's Dependence on Length Scale. (*In preparation*).

RESEARCH EXPERIENCE

Graduate Research Assistant | Boston College (08/2021 – Present)

Department of Physics & Schiller Institute for Integrated Science and Society

- Advisor: Dr. Yi Ming (08/2023 – Present) | *Geophysical Fluid Dynamics*
 - Constructed a diagnostic framework to isolate fundamental physical mechanisms which influence the tropical atmospheric circulation.
 - Evaluated the response of Hadley and Walker circulations to zonally asymmetric forcings and elevated CO₂ concentrations in an idealized general circulation model.
 - Contributed to methodology used to investigate the tropical circulation response to warming scenarios in CMIP6 model ensemble.
 - Evaluated the time-mean surface wind trends in ERA5 reanalysis product and their connection to the divergent circulations.
 - Configured and executed idealized climate simulations on high-performance computing (HPC) clusters utilizing Fortran and Python within a Unix/Linux environment.
 - Mentored undergraduate researchers in computational workflows and high-performance computing techniques.
- Advisor: Dr. Benedetta Flebus (08/2021 – 08/2023) | *Theoretical Condensed Matter Physics*
 - Evaluated Magnon-phonon interactions in non-Hermitian second-quantized systems.

Undergraduate Research Assistant | The Ohio State University (10/2016 – 05/2021)

Department of Physics

- Advisor: Dr. Ciriya Jayaprakash (04/2019 – 05/2021) | *Non-Linear Dynamical Systems*
 - Developed 2D discretized numerical simulations and 1D continuous partial differential equations (PDEs) in C++ and Mathematica to examine interface motion and domain stability of spatially extended stochastic dynamical systems.
 - Investigated accessible stable states available to the system using methods from statistical mechanics.
- Advisors: Dr. Andrew Heckler & Dr. Gregory Kilcup (04/2017 – 04/2019) | *Physics Education Research*
 - Created targeted diagnostic problem sets designed to reduce cognitive load and enhance foundational problem-solving skills in intermediate mechanics.
 - Analyzed statistical data on student estimation of physical quantities to identify correlations between physical intuition and overall academic performance.
- Advisor: Dr. Jay Gupta (10/2016 – 04/2017) | *Experimental Condensed Matter Physics*
 - Worked to construct a tabletop Raman spectrometer to conduct optical imaging and quality characterization of 2D materials and monolayer graphene.
 - Utilized Zeiss optical microscopy to evaluate micro-structural properties of synthesized graphene samples.

TALKS AND POSTERS

Conference Presentations

- **Hunt, P.**, Kim, S., MacDonald, C. G., & Ming, Y. (2025). *Convective reorganization under different zonally-asymmetric thermal forcings and CO₂ concentrations in an idealized model*. AGU Fall Meeting, New Orleans, LA.
- **Hunt, P.**, Kim, S., & Ming, Y. (2025). *Hadley circulation under different zonally-asymmetric thermal forcings and CO₂ concentrations in an idealized model*. Gordon Research Conference (GRC), Lewiston, ME.
- **Hunt, P.**, Kim, S., & Ming, Y. (2024). *Interconnectivity of tropical circulation patterns*. ERDEC Conference, Chestnut Hill, MA.

Departmental Seminars

- **Hunt, P.**, Pu, L., Kim, S., MacDonald, C. G., & Ming, Y. (2026). *Convective Reorganization*. Department of Earth and Environmental Sciences Seminar, Boston College.
- **Hunt, P.** (2026). *Topological Origins of Equatorial Waves*. Department of Physics Seminar, Boston College.

TEACHING EXPERIENCE

Graduate Teaching Assistant | Boston College (08/2021 – Present)

- Instructed lab and recitation sections across many distinct physics courses, including: Graduate Electrodynamics, Introductory Physics (Mechanics and Electromagnetism), Classical Mechanics for Engineers, Physics for Pre-Medical Students, and Physics of the Environment.
- Communicated complex physical principles to diverse student cohorts, led interactive problem-solving sessions, and provided detailed feedback on coursework.

Undergraduate Teaching Assistant | The Ohio State University (08/2018 – 05/2021)

- Facilitated flipped-classroom discussion sections for 200+ students in Intermediate Mechanics (PHYS 2300/2301) utilizing Morin's *Classical Mechanics* textbook.
- Conducted midterm review workshops for 40+ students and held individual tutoring sessions to reinforce theoretical concepts.
- Led lab and recitation sections for introductory calculus-based mechanics, adapting curriculum delivery to remote learning formats via Zoom.

TECHNICAL & COMPUTATIONAL SKILLS

- Programming Languages: Python, Fortran, C++, Mathematica, Java
- Computational & Modeling Methods: High-Performance Computing (HPC / Linux / Unix), Idealized Climate Modeling, PDE Numerical Simulations, Stochastic Systems

AWARDS & HONORS

- 3rd Place, NeurIPS Competition on Reinforcement Learning, Boston University (2022)
- James Smith Scholarship, The Ohio State University (2019)

- Undergraduate Summer Research Scholarship, The Ohio State University (2018)
- Dean's List (7 of 8 Semesters), The Ohio State University
- Freshman Physics Academic Achievement Award, The Ohio State University

ACADEMIC OUTREACH

Student Board Member | Schiller Institute for Integrated Science and Society, Boston College (2024 – Present)

- Planned and hosted informational seminars on research opportunities and student involvement across disciplines.
- Founded a service sub-committee dedicated to community service initiatives in the greater Boston area.
- Served as the graduate student representative during interviews and evaluation processes for new faculty candidates.

Event Volunteer | Cloud Feedback Model Intercomparison Project (CFMIP) Meeting (2024)

- Assisted with event coordination, technical setup, attendee check-in, and poster session management for international climate research attendees.

Event Volunteer | APS Conference for Undergraduate Women in Physics (CUWiP) (2023)

- Facilitated conference operations, handled attendee registration, managed room assignments, and directed poster session setups.

RELEVANT GRADUATE COURSEWORK

- Geophysical Modeling (Dept. of Earth & Environmental Sciences, BC)
- Mathematical Physics (Dept. of Physics, BC)
- Climate, Energy, and Sustainability Nexus (Schiller Institute, BC)
- Artificial Intelligence and Deep Learning (Dept. of Computer Science, BU)
- Big Data Analytics & AI Methods (Dept. of Physics, OSU)
- Particle Physics (Dept. of Physics, BC)
- Topology in Condensed Matter (Dept. of Physics, BC)
- Computational Physics (Dept. of Physics, OSU)
- Applied Differential Geometry in General Relativity (Dept. of Mathematics, OSU)

REFERENCES

Dr. Yi Ming

Professor, Department of Earth and Environmental Sciences
Core Faculty, Schiller Institute for Integrated Science and Society
Boston College
Relationship: Ph.D. Dissertation Advisor
Email: Ming.Yi@bc.edu

Dr. Pu Lin

Senior Research Scientist, Department of Earth and Environmental Sciences
Boston College
Relationship: Primary research collaborator & Co-Author
Email: Lin.Pu@bc.edu

Dr. Cameron MacDonald

Environmental Fellow (Postdoctoral Researcher)
Harvard University
Relationship: Research collaborator & Co-Author
Email: cgmacdona1d@fas.harvard.edu

Dr. Ciriya1m Jayap1rakash

Professor Emeritus, Department of Physics
The Ohio State University
Relationship: Undergraduate thesis advisor
Email: jay@physics.osu.edu